WHAT IF SOMEONE HACKED THE UK POWER GRID RIGHT NOW?

AI-Based Cyberattack Detection in Smart Power Grid Systems

Real-time AI detection, classification, and explanation of cyberattacks in power systems

Hei Yu Chau | 2370932 | Swansea University Prifysgol Abertawe

94.6%
ATTACK DETECTION RECALL

10,536 / 11,133 attacks caught

1. THE PROBLEM

- Modern power grids are **increasingly vulnerable** to cyberattacks.
- The 2015 Ukraine attack left **225,000 people** without electricity.
- Most AI systems are **black-box models** with no human control.

2. PROPOSED SOLUTION

Detect, classify, explain, and decide — with human verification.

Detect	Classify	Explain	Decide
Attacks in Real Time	Attack Types Accurately	Decisions with Key Features	Human-in-the-Loop Verification

AI + Explainability + Human-in-the-Loop = Safer Power Grids

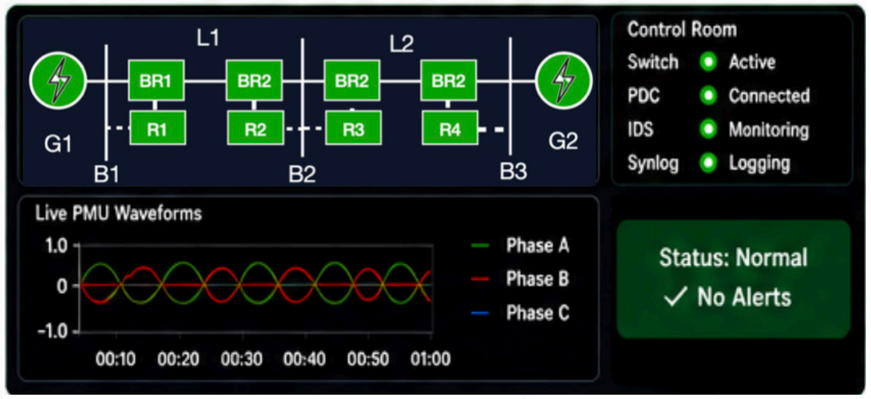
3. SYSTEM PIPELINE (End-to-End Workflow)

1. **DATA** — Real-time signals
2. **PREPARE** — Clean & normalise
3. **DETECT** — Normal vs Attack
4. **TYPE** — Attack type
5. **SCENARIO** — Exact attack
6. **DECIDE** — Human verification

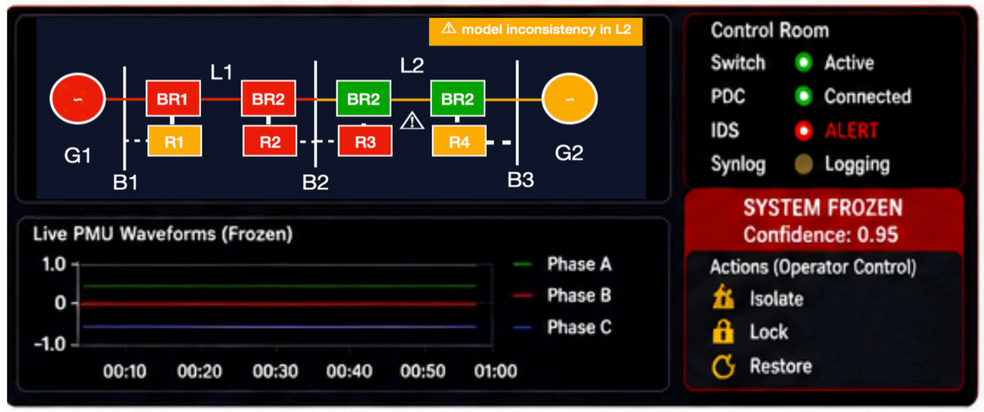
FALLBACK MECHANISM: Uncertain result → Re-check (safe route)

4. DASHBOARD OVERVIEW (Debug / Live)

NORMAL OPERATION – SYSTEM HEALTHY



ATTACK DETECTED – SYSTEM FROZEN



5. HUMAN-IN-THE-LOOP

When an attack is detected, the system **pauses**.
The operator reviews **explainable insights** before action.

Human-verified decisions ensure safety in critical infrastructure.

- Real-time alerts with **confidence scores.**
- **Explainable** outputs (key features driving decisions).
- **Decision support** for response actions (isolate, restore, ignore).
- Response time: **< 1 second** per event.

6. KEY PERFORMANCE (Test Set)

★ PRIMARY

94.6% RECALL ★
Detects nearly all attacks in real time
10,536 / 11,133 attacks caught

75.6% ACCURACY
Overall system accuracy

76.5% PRECISION
When system says attack, it's right 3 out of 4 times

7. SYSTEM FEATURES

- Real-time detection + confidence score
- Live PMU monitoring (freeze on alert)
- Grid localisation (fault source)
- Explainable AI (key features + decision path)
- Human-in-the-loop decision control
- Debug & Live modes

8. TECHNICAL NOTES

DATA: 78k | PMU/SCADA | LR + RF

TRADE-OFF: High recall | false alarms

SETUP: Simulated

LIMIT: Human delay

9. KEY INSIGHT

The real challenge is not detection — but distinguishing cyberattacks from real physical faults.

SCAN ME
Demo + Code + Full System Details